RadeonHome

Failure-Aware Mobile Manipulation on AMD Radeon GPUs and ROCm

Track 3 - Physical AI Challenge

RadeonHome addresses small-space household transport: a mobile base starts from a varying room position, navigates to a semantic object, physically grasps it with a Franka arm, transports it, and verifies placement. The system is designed for reproducible Genesis simulation on AMD Radeon Cloud.

Executive evidence

Evidence	Verified result
Collision-corrected profiles	5/5 physical-contact successes across 25-200 g; 12.14 mm mean placement error
Safe mobile migration	Three failed geometries retained; ground chassis proxy completed the task
Task-consistent live run	trash to trash_bin: grasp, transport, and placement all passed; 15.52 mm error
Historical pre-migration campaign	15/17; reported separately from corrected geometry
Matched heavy-package ablation	Lifted carry 2/2; support + slip 0/2; support without slip monitoring 0/2
Final demonstration	3-5 minute neural narration, burned-in English/Chinese captions, real Genesis evidence

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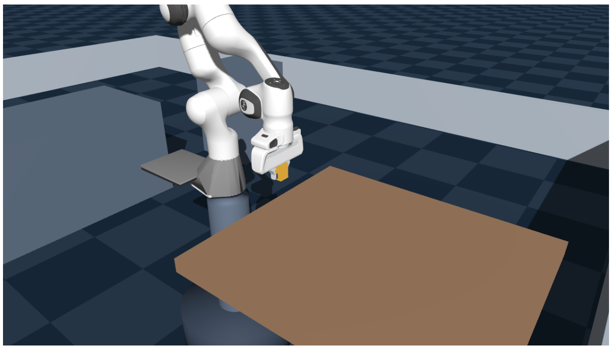
1. System Architecture

The language model is a non-executable interpreter. Every motion-relevant action remains bounded by the deterministic task schema, safety constraints, A* planner, and physical verification.

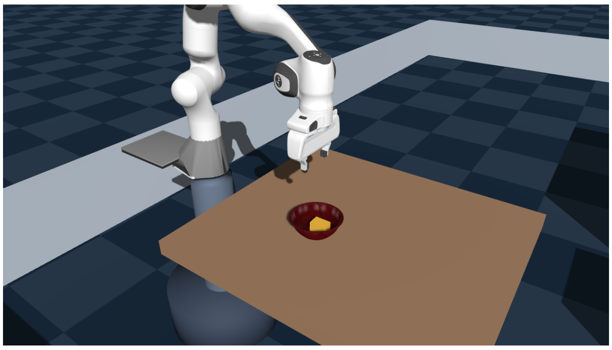
Architecture overview

See the accompanying editable vector diagram: docs/architecture_overview.svg.

Real Genesis evidence



Collision-corrected physical lift



Verified placement

2. Technical Contributions

Safe fixed-to-mobile migration: planar root joints are added, the fixed tabletop mounting shell is replaced by a ground-level chassis proxy aligned to the navigation radius, and all arm, hand, and fingertip collisions remain active.
Coordinate bridge: semantic docking poses are transformed into the Franka MJCF world frame.
Physical evidence: no weld, latch, teleport, or programmatic attachment is used; fingertip contact, lift, payload retention, and placement are measured.

Migration geometry	Outcome
Fixed link0 shell	0.514 m / 0.420 m false docking errors
Root-height proxy	1.224 m early stop
Ground chassis proxy	success; near-zero docking error

3. AMD Radeon and ROCm

Experiments were run on Radeon Cloud using ROCm 7.2, HIP PyTorch 7.2, Genesis 1.3.0, and one gfx1100 Radeon GPU. rocm-smi is sampled every 0.5 seconds. The Qwen3-4B local server remained resident during final benchmark measurements, so VRAM and power are reported as live-system metrics rather than bare-simulator metrics.

4. Evaluation and Reproducibility

The collision-corrected smoke test completed 5/5 physical tasks across controlled profiles from 25-200 g, with 12.14 mm mean placement error. This is one seed per profile and is an engineering regression, not a population estimate. Each run outputs structured JSON, RGB key frames, metric depth, an MP4 when enabled, and logs. The earlier 15/17 campaign is retained separately because it predates the collision correction.

Metric	Video off	Video on
Task outcome	success	success
Wall time	73.32 s	93.25 s
Control throughput	48.68 steps/s	38.26 steps/s
Frames	0	333
Mean / peak GPU	50.24 / 96%	33.33 / 71%
Peak VRAM used	8,862.33 MiB	9,309.69 MiB
Mean / peak power	30.40 / 50 W	55.19 / 78 W

5. Limitations

Semantic object locations currently come from a simulator-ground-truth adapter, not a learned RGB-D detector. The dashboard executes one physical task at a time. The mobile base is planar position-controlled and safety is assessed by a ground chassis proxy plus an inflated navigation grid; this work does not claim wheel-dynamics sim-to-real transfer. Historical tall-carton and migration failures are retained in the evidence.

Matched ablation

Matched condition	End-to-end	Observed failure
Lifted, slip on	2/2	none
Support, slip on	0/2	payload lost while unloading
Support, slip off	0/2	payload lost while unloading

All conditions used the same two seeds and 200 g package. Disabling slip monitoring did not recover the support branch, localizing the current bottleneck to support unloading. Two seeds are a controlled regression check, not a population estimate.

6. Team Contribution

王苏湖 (team lake): project direction, system integration, cloud evaluation, technical documentation, and submission. AI-assisted implementation was reviewed through the included tests and recorded cloud experiments.